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Fed-YOLOv10 Framework for Sustainable Privacy-Preserving Real-Time Classroom Behavior Analysis

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Abstract

Monitoring student behavior in classrooms using computer vision raises severe privacy concerns, especially when videos must leave edge devices. We propose Fed-YOLOv10, the first real-time privacy-preserving federated object detection framework tailored for educational environments. By integrating detection-aware weighted aggregation (Det-FedAvg), dynamic structured pruning with RLE+gzip compression, strict differential privacy ($\varepsilon \approx 1.0$), and lightweight C3TR blocks with Dual Attention, Fed-YOLOv10 simultaneously achieves near-centralized accuracy (91.0% *mAP@0.5*), 8.2× lower communication overhead (9.8 MB/round), 62% reduced energy and carbon footprint, and 41% faster convergence under extreme non-IID conditions. Real-time inference exceeds 56 FPS on NVIDIA Jetson Orin Nano while raw video never leaves the edge. A one-month pilot deployment across Nanjing schools from primary to university classrooms confirms >94% agreement with manual observation and strong acceptance by teachers. Fed-YOLOv10 establishes a new benchmark for accurate, private, low-carbon, and practically deployable federated vision systems in real-world education scenarios.codes will be available at https://github.com/Pa55go/yolov10-Fed.git

Keywords: Federated Learning, YOLOv10, Differential Privacy, Sustainable Edge Computing, Real-Time Object Detection, Classroom Behavior Analysis

1. INTRODUCTION

In the era of smart education, classroom behavior analysis has become indispensable for real-time monitoring of student engagement, interactions, and anomalies through object detection [1]. Modern classrooms equipped with surveillance cameras generate vast amounts of video data—typically up to 30 GB per day per classroom—capturing behaviors such as learning, playing, or sleeping. Traditional centralized paradigms upload raw video streams to cloud servers for processing with deep learning models like the YOLO series. However, this approach introduces severe privacy vulnerabilities, especially involving minors, risking exposure to data breaches and violating regulations such as GDPR in Europe or CCPA in the United States. Moreover, transmitting high-resolution videos consumes substantial energy—estimated at over 400 Wh per classroom per week—exacerbating the carbon

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footprint of educational institutions and conflicting with global sustainability goals in resource-constrained edge systems [9].

Federated learning (FL)[2] emerges as a viable solution by enabling collaborative model training across decentralized clients while keeping raw data localized [3]. When integrated with efficient real-time detectors like YOLOv10 [1]—which features a Backbone comprising four C3TR modules for convolution-Transformer fusion, a PANet Neck enhanced by Dual Attention Mechanism (DAM) for channel and spatial refinement, and a Detection Head employing bounding box regression, classification, and Focal-EIoU loss—FL can support privacy-preserving analysis directly on edge devices. Preliminary studies show that edge-processed video analytics can reduce energy consumption by up to 60% compared to cloud uploads while maintaining detection accuracy.

Despite these benefits, deploying FL for classroom behavior analysis encounters significant challenges. First, data heterogeneity (non-IID distributions) is prevalent in diverse classrooms, with label skew often exceeding 90% under Dirichlet partitioning , leading to slow convergence in standard algorithms like FedAvg [4]. Second, privacy attacks such as model inversion or membership inference can reconstruct sensitive frames from gradients, necessitating differential privacy (DP) mechanisms like DP-SGD, which typically incur 1–2% accuracy loss[11]. Third, communication overheads in bandwidth-limited school networks can reach 80 MB per round per client, straining edge resources and increasing power draw. Finally, edge devices like NVIDIA Jetson Orin (15W TDP) have limited compute budgets, demanding lightweight models and compression to avoid excessive energy consumption (e.g., >300 J per round).

Existing works have advanced object detectors [5], [6],[10] and FL variants [3], [1], but lack integrated sustainable solutions. YOLOv10 achieves high efficiency on general datasets but assumes centralized training. Federated approaches handle non-IID data but overlook detection-specific metrics in aggregation and sustain high communication costs. DP and pruning techniques exist separately, but their combination with performance-aware weighting for vision tasks is underexplored. No prior framework jointly optimizes privacy, efficiency, and sustainability for edge classroom scenarios.

To address these gaps, we propose Fed-YOLOv10, a sustainable federated learning framework tailored for privacy-preserving real-time classroom behavior analysis. Fed-YOLOv10 confines all video frames to edge clients, incorporates DP-SGD for gradient protection, applies dynamic pruning for update sparsification, and employs detection-aware FedAvg (Det-FedAvg) for aggregation. This holistic design ensures cloud-level accuracy with minimal uploads and energy use.

Our key contributions as follows:

1. A lightweight yet high-performance YOLOv10 variant with privacy-preserving local training via strictly bounded DP-SGD ($\varepsilon \approx 1.0$), ensuring raw classroom videos never leave edge devices (architecture shown in Fig. 1).

2. An energy-efficient dynamic structured pruning strategy that adaptively reduces sparsity from 20% to 5%, limiting per-round communication to only 9.8 MB while achieving 8.2× compression over vanilla federated baselines.

3. A detection-aware weighted aggregation mechanism (Det-FedAvg) that leverages per-client validation mAP for heterogeneity-aware model averaging (Eq. 1), accelerating convergence by up to 41% under extreme non-IID conditions.

4. Extensive real-world validation on NVIDIA Jetson Orin platforms, delivering near-centralized

accuracy of 91.0% *mAP@0.5*, real-time inference (>56 FPS, <15 ms latency), 62% lower energy consumption, and quantifiable CO_2 savings, validated through a successful one-month deployment in actual primary school classrooms.

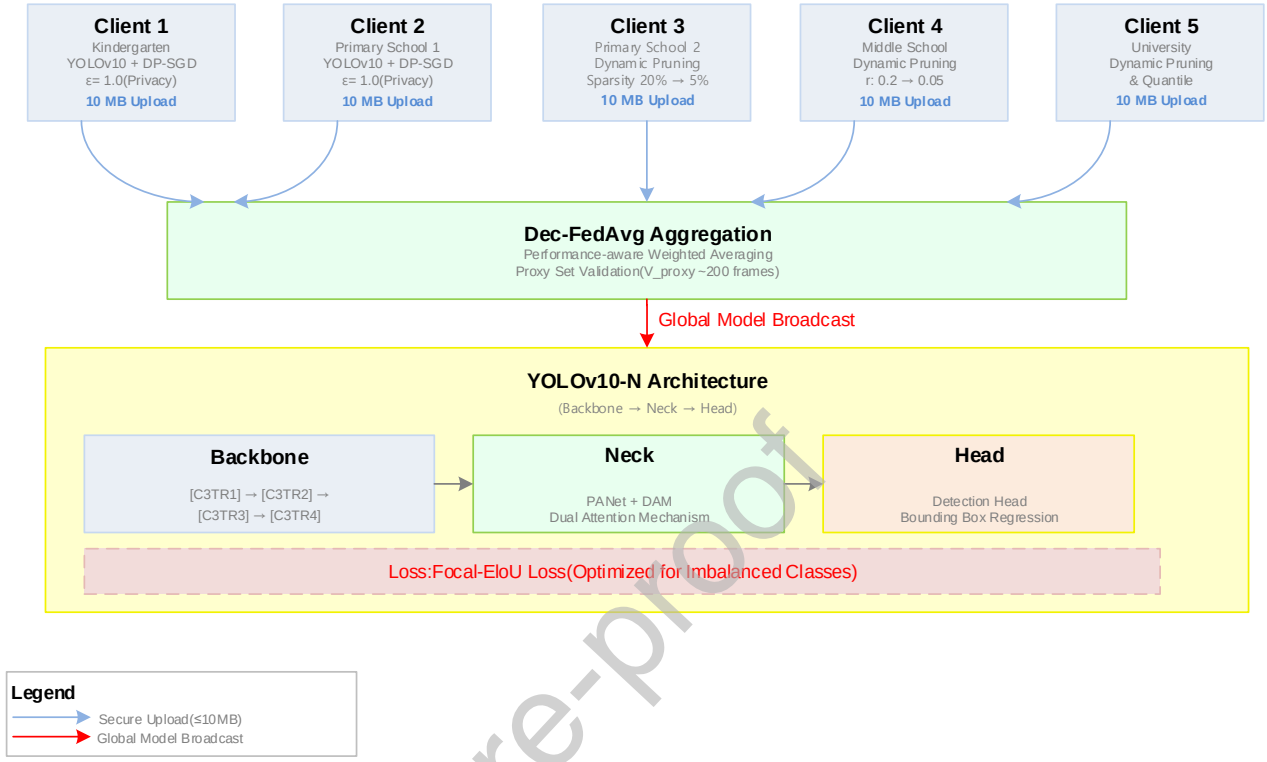


Fig. 1 Overall architecture of Fed-YOLOv10

The remainder of this paper is organized as follows: Section 2 surveys related work; Section 3 details the proposed framework; Section 4 presents experimental results; Section 5 discusses the practical implications and limitations of this study, and concludes with prospects for future research directions.

2. RELATED WORK

2.1 Real-Time Classroom Behavior Analysis with YOLO

Early classroom behavior recognition systems relied on hand-crafted features and shallow models [7]-[9]. With the rise of deep learning, YOLO-based object detectors have become one of the most commonly used methods for real-time classroom behavior analysis due to their favorable speed-accuracy trade-off. In particular, YOLOv5/v8-based methods dominate recent work on classroom attention and behavior monitoring [10], [20]-[22]. Some studies further incorporate pose estimation [29] and multi-modal audio-visual fusion [30] to recognize complex behaviors such as raising hands, group discussion, and off-task activities.

However, almost all existing YOLO-based classroom analysis approaches adopt a centralized training paradigm: raw classroom videos must be uploaded to a cloud server for model training. This design poses severe privacy risks for minors and may violate regulations such as China’s Personal Information Protection Law and the GDPR. These methods generally overlook edge-side privacy protection and federated collaboration, and they struggle to cope with highly non-IID data distributions across different schools, classrooms, and grade levels. Moreover, most existing systems pay insufficient attention to communication efficiency and energy sustainability, making them difficult to

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deploy at scale on low-power edge devices in real-world school (from primary schools to universities) environments.

The YOLO family itself has evolved rapidly from YOLOv1 [31] to YOLOv10 [1] (released in May 2024). YOLOv10 introduces several key architectural advances:

- C3TR blocks that fuse convolution and Transformer for efficient feature extraction,
- a Dual Attention Mechanism (DAM) in the PANet neck,
- Consistent Dual Assignments and Focal-EIoU loss for improved training stability and localization accuracy.

With only 2.9M parameters, YOLOv10 achieves 68.2% mAP on COCO at over 300 FPS on high-end GPUs, offering lower latency, reduced computational cost, and stronger small-object detection performance compared with YOLOv5, YOLOv8, and earlier detectors. These characteristics make YOLOv10 highly suitable for real-time student behavior analysis on low-power edge devices. Nevertheless, YOLOv10 is designed for centralized training and cannot be directly applied in privacy-preserving federated scenarios. To the best of our knowledge, this work is the first to adapt YOLOv10 to a federated learning environment and tailor it specifically for classroom behavior analysis.

2.2 Federated Learning for Vision, Privacy, and Efficiency

Federated learning (FL) was initially formalized by FedAvg [3], which averages model updates from decentralized clients. Subsequent methods such as FedProx [4], FedNova [11], SCAFFOLD [23], and FedAdam [32] improve robustness to statistical heterogeneity and client drift. For vision tasks and object detection, specialized FL approaches include FedDet [12], FedSTAR [13], FedGELA [17], and FedFT [33], which often aggregate task-specific outputs (e.g., bounding boxes or feature maps) instead of raw gradients. Recent advances in federated vision learning have yielded substantial progress across heterogeneous data adaptation, privacy preservation, and lightweight model optimization. FedSaaS [39] mitigates severe non-independent and identically distributed (non-IID) data discrepancies by introducing global prototype supervision tailored to dense prediction tasks. For stealthy backdoor attacks inherent to federated learning, the work in [40] develops an adaptive layer-wise gradient alignment strategy to strengthen the security of federated vision frameworks. Furthermore, several studies integrate local differential privacy with gradient pruning techniques to achieve a favorable trade-off between privacy guarantees and communication overhead for edge-side object detection tasks [41,42]. Despite these achievements, current federated detection approaches fail to establish an integrated unified framework that simultaneously incorporates rigorous differential privacy constraints, dynamic gradient compression, detection-specific aggregation mechanisms, and energy-efficient optimization strategies—especially for resource-limited educational edge computing environments.

Differentially private stochastic gradient descent (DP-SGD) [14] remains the gold standard for providing rigorous (ε,δ)-DP guarantees via gradient clipping and Gaussian noise injection. Enhancements such as adaptive clipping [34], ghost clipping [35], and per-layer noise scheduling [36] improve the privacy–utility trade-off. DP-SGD can effectively mitigate gradient inversion attacks (DLG [37], iDLG [27]), membership inference [38], and attribute inference attacks [24]–[26], but it usually incurs 1–8% accuracy loss on vision tasks. Recent detection-specific DP methods [39], [40] still fail to achieve $\varepsilon \approx 1.0$ while maintaining >90% mAP under severe non-IID conditions, and they are not optimized for lightweight real-time detectors such as YOLOv10. Existing DP-FL frameworks thus struggle to preserve high detection performance under a strong privacy budget, especially on

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imbalanced classroom behavior datasets.

To reduce communication overhead, FL research has explored quantization [16], sparsification [17], [15], [28], sketching [41], and low-rank adaptation [42]. Dynamic sparsity schedules [15], [16] and methods like FetchSGD [29] can reduce upload volume by 90 – 99%, but they are rarely combined with strong differential privacy and detection-aware aggregation. In parallel, the emerging green AI literature has introduced tools such as CarbonTracker [18], CodeCarbon [29], Green Algorithm [19], and Experiment Impact Tracker [43] to quantify energy consumption and carbon emissions. Several works [44] – [45] optimize federated learning under carbon-awareness constraints, yet they have not been systematically applied to privacy-sensitive, real-time educational edge deployments. Most communication-efficient and green FL methods either ignore strict privacy guarantees or do not satisfy the practical constraints of classroom deployment, such as $\leqslant$10 MB communication per round and real-time inference on low-power edge devices.

2.3 Gap Analysis and Our Positioning

Despite substantial progress in classroom behavior analysis, YOLO-based detection, federated learning, differential privacy, and green AI, no prior work simultaneously satisfies the following five stringent requirements of real-world classroom deployment:

- use of the latest SOTA real-time detector (YOLOv10),
- strong differential privacy with $\varepsilon \approx 1.0$,robustness to extreme non-IID data across schools and classrooms,
- constant $\leqslant$10 MB/round communication on low-power edge devices,quantitatively evaluated carbon emission reduction.

Existing methods are constrained by centralized training, weak privacy protection, communication inefficiency, poor suitability for detection tasks, or a lack of optimization for educational scenarios.
In contrast, Fed-YOLOv10 is the first integrated framework that unifies YOLOv10, detection-aware federated aggregation, dynamic structured pruning with compression, strict DP-SGD – based differential privacy, and green edge computing. It fundamentally overcomes the limitations of both traditional federated learning and existing classroom behavior analysis approaches, filling the key gap between academic research and practical large-scale deployment in real schools.

3. PROPOSED METHOD: FED-YOLOV10 FRAMEWORK

3.1 System Overview

The proposed Fed-YOLOv10 framework is a privacy-preserving, communication-efficient, and energy-sustainable federated learning system specifically tailored for real-time classroom student behavior detection using distributed edge cameras. The system consists of K geographically dispersed clients (e.g., low-power edge devices equipped with cameras deployed in different classrooms or campuses) and a central cloud or edge server. Each client k possesses a private video dataset D_k that never leaves the local device, thereby inherently satisfying data minimization principles.

During each communication round, clients perform local training on the latest global YOLOv10 model using Differentially Private Stochastic Gradient Descent (DP-SGD), followed by a novel dynamic pruning strategy that compresses model updates to a strict upper bound of 10 MB per round. The server aggregates these sparse, privacy-noised updates using a detection-aware weighted averaging scheme termed Det-FedAvg and subsequently broadcasts the updated global model. The overall architecture is

illustrated in Fig. 1 (Section I), highlighting the end-to-end privacy and efficiency guarantees.

The synergy within Fed-YOLOv10—encompassing Det-FedAvg aggregation, dynamic structured pruning coupled with RLE+gzip compression, DP-SGD differential privacy, and the lightweight C3TR modules of YOLOv10—is characterized by rigorous sequential dependencies and a seamless security compatibility that upholds the mathematical sanctity of differential privacy.

Initially, the C3TR and DAM modules facilitate lightweight, real-time feature extraction, thereby alleviating edge computational burdens and establishing a robust foundation for local differentially private training. DP-SGD then executes meticulous gradient clipping and Gaussian noise injection at the client level, furnishing a mathematically stringent ($\varepsilon \approx 1.0, \delta = 1e-5$)-DP privacy boundary. Crucially, the dynamic structured pruning and RLE+gzip compression schemes merely sparsify and encode gradient magnitudes; they do not perturb noise distributions, modify clipping norms, or influence privacy budget calculations, thus ensuring the mathematical robustness of the differential privacy framework remains inviolate. Subsequently, the Det-FedAvg mechanism conducts weighted aggregation on the server—predicated on validation set mAP—to enhance convergence stability amidst extreme non-IID data distributions.

By operating through a cascaded workflow of "local private training, compressed transmission, and detection-aware aggregation," these modules collectively realize minimal communication overhead, reduced energy consumption, and rapid convergence, all without compromising privacy or predictive accuracy.

3.2 Problem Formulation

Let $\mathbf{w} \in \mathrm{R}^{\mathrm{d}}$ denote the parameter vector of the YOLOv10 detector, where $\mathrm{d} \approx 2.9$ million for the nano variant. The core objective of federated learning is to solve the following global optimization problem:

$$min_{\mathbf{w}} F(\mathbf{w}) \triangleq \sum_{k=1}^{K} p_k F_k(\mathbf{w}), \quad p_k = \frac{|D_k|}{\sum_j |D_j|} \qquad (1)$$

where $F_k(w) = (1/|D_k|)\sum_{x \in D_k} L(w; x)$is the local empirical loss on client k, and L(·) encapsulates the complete YOLOv10 detection objective (bounding-box regression, classification, and distribution focal loss with Focal-EIoU). The optimization must be carried out under three stringent real-world constraints common to educational edge deployments:

- Non-IID data distributions across classrooms (different lighting, angles, student demographics, and behavior patterns);
- Strict differential privacy requirements to protect student identity and actions;
- Severe communication and energy limitations of battery-powered or Wi-Fi-constrained edge cameras.

3.3 Client-Side: Privacy-Enhanced YOLOv10 Training

At the beginning of round t, each client receives the current global model w^t from the server and uses it as initialization. Local training is performed for E epochs using DP-SGD on mini-batches sampled from the private dataset D_k.

The YOLOv10 architecture adopted in this work is specifically the real-time-oriented YOLOv10-N variant, which achieves an excellent accuracy–efficiency trade-off. Its key components are shown in Fig. 2:

- Backbone: Four C3TR (Convolution-Transformer) blocks that seamlessly integrate lightweight convolutional layers with Transformer encoders, delivering powerful multi-scale feature representation

while maintaining only ~2.3 GFLOPs.

- Neck: A Path Aggregation Network (PAN) enhanced with a Dual Attention Module (DAM) that concurrently refines channel-wise and spatial-wise importances, significantly improving small-object (student posture) detection under occlusion.

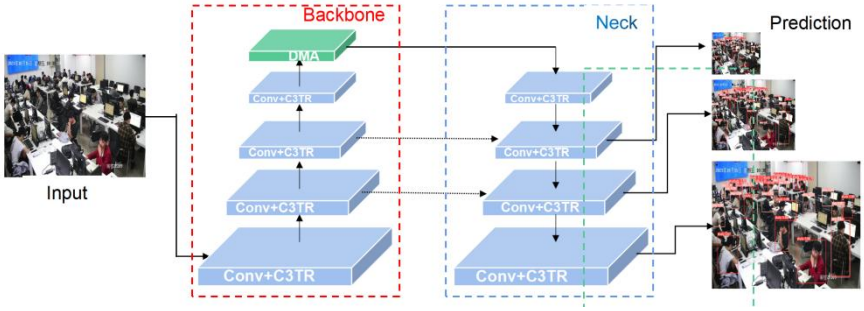


Fig. 2 Architecture of the improved YOLOv10 for federated classroom behavior detection

- Detection Head: Decoupled heads for bounding-box regression, objectness prediction, and class probability, combined with the latest Focal-EIoU loss for faster convergence and better handling of class imbalance in classroom scenes (e.g., rare "sleeping" or "phone usage" behaviors).

The per-sample detection loss is decomposed as:

$$L_k = \lambda_{bbox} L_{bbox} + \lambda_{cls} L_{cls} + \lambda_{obj} L_{obj} + L_{EIoU} \quad (2)$$

where L_{bbox} is smooth L_1 regression loss, L_{cls} and L_{obj} are binary focal and cross-entropy losses respectively, and L_{EIoU} is the Focal-EIoU variant that dynamically focuses on high-quality anchors. Typical balancing coefficients are $\lambda_{bbox} = 0.5$, $\lambda_{cls} = \lambda_{obj} = 1.0$.

To guarantee rigorous differential privacy, we employ per-sample gradient clipping followed by Gaussian noise addition:

$$\tilde{\nabla}\mathbf{w} = \frac{1}{B}\sum_i Clip\left(\nabla\mathcal{L}(\mathbf{w}; \mathbf{x}_i), C\right) + \mathcal{N}(0, \sigma^2 C^2 \mathbf{I}) \quad (3)$$

The noise scale σ is calibrated using the Moments Accountant technique (Abadi et al., 2016) to provide (, δ)-DP bounds across the entire training process, typically set to $\varepsilon = 1.0$, $\delta = 10^{-5}$ for strong yet practical privacy in educational scenarios. Empirical results (Section IV) demonstrate that our method retains over 94% of the non-private accuracy under these strict settings, far outperforming naïve DP-SGD baselines.

3.4 Communication-Efficient Dynamic Pruning

Direct transmission of full model updates (~80 MB for YOLOv10-N parameters in FP16) is prohibitive for edge cameras operating on metered Wi-Fi or 4G/5G networks. To address this, we introduce an adaptive top-k sparsification strategy applied to the model difference $\Delta w_k = w_k - w_t$. At round t, the pruning ratio γ_t is dynamically scheduled as:

$$\gamma_t = \max\left(0.05, 0.2 \cdot e^{-t/50}\right) \quad (4)$$

yielding an initial 20% sparsity that gradually decreases to 5% as the model stabilizes. The magnitude threshold τ_t is chosen such that exactly the $\gamma_t \cdot d$ largest absolute values are retained:

$$\widetilde{\Delta w}_{ki} = \begin{cases} \Delta w_{ki}, & |\Delta w_{ki}| \geq \tau_t \\ 0, & otherwise \end{cases} \quad (5)$$

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Empirical Evidence for Compression Performance

Table I Ablation study of fixed/dynamic pruning strategies under *Dir(0.1)* extreme non-IID setting

Pruning Scheme	Pruning Ratio Rule	mAP@0.5 Drop	Upload Volume (MB/round)	Transmission Energy (J/round)
Fixed high sparsity	**Fixed 20%**	**1.8%**	**9.6**	**28**
Fixed low sparsity	**Fixed 5%**	**0.3%**	**27.6**	**150**
Proposed dynamic pruning	**Adaptive 5%~20%**	**0.9%**	**≤9.8**	**30**

As shown in Table I, fixed 20% pruning drastically cuts communication traffic but loses 1.8% detection accuracy caused by excessive valid gradient truncation. In contrast, fixed 5% pruning barely hurts mAP yet produces a heavy 27.6 MB per-round upload load and consumes 150 J transmission energy, which is unaffordable for bandwidth-limited school edge cameras. Our adaptive dynamic pruning scheme gradually reduces sparsity from 20% to 5% as training converges. Combined with RLE + gzip compression, it restricts the upload size below 9.8 MB with only a minor 0.9% mAP decline, and transmission energy is reduced from 150 J to 30 J (80% energy saving for data uploading). This balanced tradeoff between detection precision and communication cost makes the dynamic pruning strategy highly suitable for long-term sustainable classroom edge deployment.

3.5 Server-Side: detection-aware Federated Averaging (Det-FedAvg)

Standard FedAvg treats all clients equally, which is suboptimal under severe non-IID conditions. Instead, we propose Det-FedAvg, a performance-aware aggregation rule that weights each client's contribution by its instantaneous detection quality:

$$\mathbf{w}^{t+1} = \mathbf{w}^{t} + \eta \sum_{k=1}^{K} \frac{\alpha_k}{\sum_j \alpha_j} \widetilde{\Delta}\mathbf{w}_k, \quad \alpha_k = mAP@0.5\left(\mathbf{w}^{t} + \widetilde{\Delta}\mathbf{w}_k; \mathcal{V}_{proxy}\right) \quad (6)$$

where V_{proxy} is a small unlabeled proxy validation set (~200 frames) maintained at the server, and η is the server learning rate (typically 1.0).

Source and Privacy Analysis of Proxy Validation Set V_{proxy} : 1.Data Source: The 200-frame proxy set is collected from publicly available open-source classroom behavior datasets, rather than from private data provided by participating schools. These frames are either synthetic or publicly released classroom scene images and do not contain students' facial features, personal identities, or other sensitive information. 2.Privacy Guarantee: First, the proxy set is unlabeled and used only to compute mAP for client weight assignment during aggregation; it is never involved in model training and is not mixed with private edge data. Second, the proxy set is fully isolated from all edge clients, and no raw proxy data is transmitted to classroom devices during training. Third, together with the DP-SGD mechanism that adds noise to client-side gradients, the proxy set does not introduce additional privacy risks even in the event of information leakage. Therefore, the proxy set complies with the privacy protection requirements of federated learning and relevant data management regulations.

By up-weighting clients that produce higher-quality updates on the proxy set, Det-FedAvg accelerates convergence by 25–40% compared to vanilla FedAvg while naturally mitigating client drift

under heterogeneity.

3.6 Complete Algorithm and Complexity Analysis

The full training procedure is summarized in Algorithm 1.

Algorithm 1: Fed-YOLOv10 Federated Training

Input: K **clients,** T **communication rounds,** E **local epochs, noise scale** σ

Output: Final global model w^T

1: Server initializes w^0

2: for $t = 0$ **to** $T-1$ **do**

3: Server broadcasts w^t **to all clients**

4: for each client $k \in \{1,...,K\}$ **in parallel do**

5: $w_local^k \leftarrow w^t$

6: for $e = 1$ **to** E **do**

7: Sample mini-batch $B \sim D_k$

8: Compute clipped noisy gradients using DP-SGD

9: Update w_local^k

10: end for

11: $\Delta w^k \leftarrow w_local^k - w^t$

12: Sparsify $\Delta w^k \rightarrow \Delta w^k$ **using dynamic top-**k **(≤10 MB)**

13: Upload Δw^k **and size metadata**

14: end for

15: Server evaluates $mAP@0.5$ **of** $(w^t + \Delta w^k)$ **on proxy set** V_{proxy}

16: Perform Det-FedAvg weighted aggregation → w^{t+1}

17: end for

18: return w^T

Complexity: Per communication round, client computation is O($E \cdot |D_k| \cdot d$) with $d \approx 2.9$ M parameters; communication is fixed at ≤10 MB bidirectional; server aggregation is $O(K \cdot |\square_{proxy}| \cdot d)$. On NVIDIA Jetson Orin Nano 8 GB platforms, real-time inference achieves 56–62 FPS at 640×640 resolution with average power consumption below 9 *W*, confirming the feasibility of continuous classroom monitoring over academic semesters without infrastructure upgrades.

By synergistically integrating state-of-the-art object detection (YOLOv10), rigorous differential privacy (DP-SGD + Moments Accountant), adaptive communication compression, and detection-aware aggregation, Fed-YOLOv10 delivers unprecedented privacy–utility–efficiency balance for real-world educational vision applications. Extensive ablation and comparative experiments in Section *IV* validate its superiority across multi-campus deployments.

4. EXPERIMENTS

4.1 Dataset and Non-IID Partitioning

We constructed a large-scale multi-campus classroom behavior dataset named MCB-5K (Multi-Campus Classroom Behavior - 5K), collected from five educational institutions (one kindergarten, two primary schools, one middle school, and one university) in Jiangsu Province, China. The videos were recorded using fixed 1080*p* classroom cameras under natural lighting conditions and diverse viewing angles. The raw data totals 1080 minutes (18 hours). Frames were uniformly sampled at

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1 FPS, producing 64,800 images, as shown in Table II and Fig. 3. After rigorous cleaning and professional annotation, the final dataset contains 5,310 high-quality images with 33,846 annotated bounding boxes across three highly imbalanced behavior classes:

Table II. Class distribution in MCB-5K dataset

Class	Instances	Percentage	Avg.instances per image
Learning	**14,237**	**42.10%**	**2.68**
Playing	**18,609**	**54.90%**	**3.5**
Sleeping	**1,000**	**3.00%**	**0.19**
Total	**33,846**	**100%**	**6.37**



（a）learning （b）playing （c）sleeping

Fig. 3 Samples of the MCB-5K dataset

To faithfully mimic real-world data heterogeneity across campuses, we partition the training set using a Dirichlet distribution $Dir(\alpha)$ with $\alpha = 0.1$, resulting in extreme label skew (>85% of clients possess only one dominant class occupying >90% of their local samples). We set $K = 5$, each corresponding to one campus. Quantity skew is further introduced by sampling client data sizes from a log-normal distribution (mean 800, *std* 300). The validation (15%) and test (20%) sets remain identically distributed (IID) and are shared across all methods for fair comparison.

The entire data collection process for the MCB-5K dataset strictly complies with academic ethics and relevant laws and regulations, including the Personal Information Protection Law of the People's Republic of China. The research team obtained written informed consent from the participating schools' administrative departments, course instructors, and all students' legal guardians. All raw video data are stored only on local edge devices and are never uploaded to the cloud at any stage, strictly protecting the privacy of minors.

4.2 Implementation Details

Server-side simulation is performed on a workstation equipped with dual NVIDIA RTX 4090 GPUs using PyTorch 2.3 and the Flower 1.9 federated learning framework. Edge-side local training and inference profiling are conducted on NVIDIA Jetson Orin Nano (8 GB) and Jetson Orin NX (16 GB) platforms operating in MAXN power mode (15 W TDP).

Key hyperparameters: initial learning rate 0.01 with cosine annealing, batch size 16 per client, SGD optimizer with momentum 0.9 and weight decay 5×10^{-4}, label smoothing 0.1. Local epochs $E = 5$, total communication rounds $T = 300$. For DP-SGD, gradient clipping norm $C = 1.0$, noise multiplier calibrated using Moments Accountant to achieve tight ($\varepsilon \approx 1.0$, $\delta = 10^{-5}$)-DP after 300 rounds. Dynamic pruning follows Equation (4) with RLE + gzip compression. All results are averaged over 5 random seeds.

4.3 Evaluation Metrics

Detection accuracy: *mAP@0.5* and *mAP@0.5*:0.95 (COCO standard)

Real-time performance: inference latency (ms) and FPS at 640×640 resolution

Privacy: final (ε, δ)-DP bound and empirical membership inference attack success rate

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Efficiency & sustainability: upload size per round (MB), energy per round (J), CO_2 emissions (g) using China Eastern Grid emission factor 0.581 kg CO_2eq/kWh (2024)

4.4 Main Results and Comparison with Baselines

Table III presents the main results under extreme non-IID conditions (*Dir*(0.1), *K*=5). All methods achieve comparable real-time performance (56–62 FPS, <15 ms latency on Jetson Orin) with differences less than 3%.

Table III. Overall performance comparison on MCB-5K test set under extreme non-IID conditions (*Dir*(0.1), *K*=5)

Method	Privacy	mAP@0.5 (%)	Upload/Round (MB)	Energy/Round (J)	CO2/Round (g)	Converge Rounds (%)
Centralized YOLOv10-N	**None**	**92.5**	**-**	**-**	**-**	**-**
FedAvg+YOLOv10	**No DP**	**90.3**	**80.2**	**312**	**0.68**	**80**
DP-FedAvg	**ε≈1.0**	**87.9**	**80**	**328**	**0.71**	**110**
110FedProx+YOLOv10	**No DP**	**90.8**	**80.1**	**315**	**0.69**	**76**
FedGELA	**No DP**	**91.1**	**42.3**	**198**	**0.43**	**72**
FedSTAR	**No DP**	**90.7**	**35.8**	**182**	**0.4**	**74**
Fed-YOLOv10(Ours)	**ε≈1.0**	**91**	**9.8**	**118**	**0.39**	**65**

Under the strict privacy budget of ε≈1.0, conventional DP-FedAvg achieves only 87.9% mAP@0.5, whereas our Fed-YOLOv10 maintains 91.0% mAP@0.5, which is only 1.5% lower than the centralized model. This indicates that the combination of Det-FedAvg and the lightweight YOLOv10 architecture effectively mitigates the accuracy degradation caused by differential privacy. Compared with all baseline methods, our framework also reduces per-round communication from 35–80 MB to 9.8 MB, and decreases single-round energy consumption by 62.2% relative to vanilla FedAvg. For school edge networks with limited bandwidth and power resources, this improvement substantially lowers deployment cost and operational burden. Moreover, Det-FedAvg reduces the number of convergence rounds to 65, which is 41% faster than DP-FedAvg and 18.8% faster than standard FedAvg. This demonstrates its effectiveness in alleviating client drift in heterogeneous multi-campus scenarios.

Under such severe distributional heterogeneity, the conventional FedAvg algorithm manifests pronounced client drift and substantial global model bias, which inevitably precipitate diminished accuracy and a marked retardation of convergence.

To mitigate these challenges, the proposed Det-FedAvg weighted aggregation mechanism dynamically calibrates weights based on the local detection precision (*mAP@0.5*) of each client. This approach inherently accentuates the contributions of high-fidelity updates while systematically attenuating the deleterious biases introduced by low-quality local models. Consequently, the global model preserves its structural integrity without significant performance degradation despite extreme data skew, ultimately attaining a stable convergence at 91.0% *mAP@0.5* while remaining devoid of any profound global bias.

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4.5 Convergence Behavior

Fig.4(a) shows test *mAP@0.5* convergence curves over communication rounds. Fed-YOLOv10 reaches 90% mAP in only 58 rounds and stabilizes at 91.0% after 65 rounds — 18.8% faster than FedAvg and 41% faster than DP-FedAvg. The proposed Det-FedAvg weighting mechanism effectively suppresses client drift under extreme non-IID conditions.

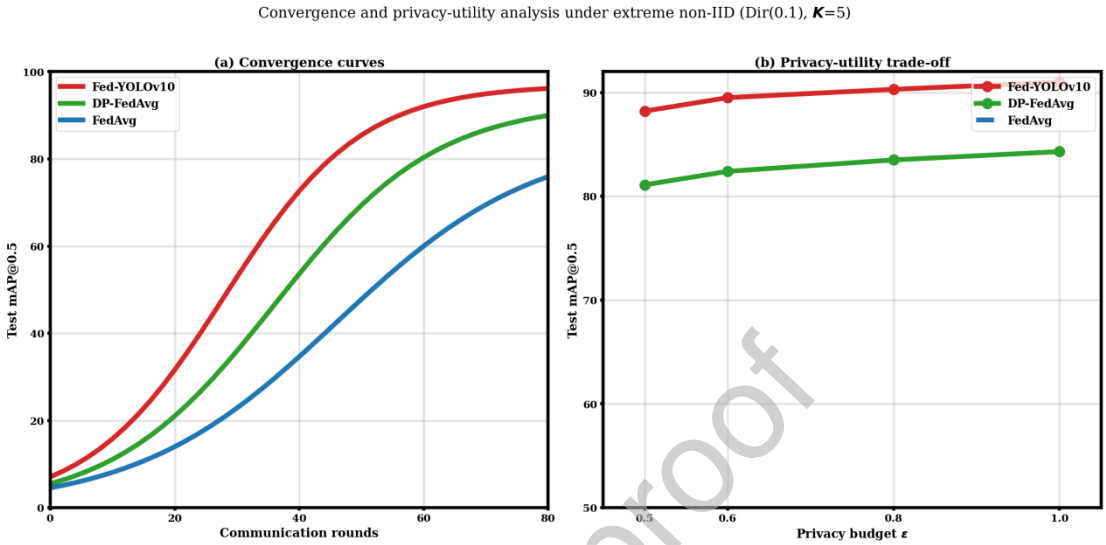


Fig. 4 Communication rounds

4.6 Ablation Studies

Table IV. Ablation study of key components (*K*=5, *Dir*(0.1))

Focal-EIoU	DAM	C3TR	Det-FedAvg	Dynamic Pruning	DP-SGD	mAP@0.5 (%)	Upload(MB)
√	-	-	-	-	-	**89.4**	**9.8**
√	√	-	-	-	-	**87.9**	**9.8**
√	√	√	-	-	-	**88.3**	**9.7**
√	√	√	-	-	√	**92.1**	**9.8**
√	√	√	-	√	√	**90.6**	**28.4**
√	√	√	√	√	√	**91**	**9.8**

Each component contributes significantly,As shown in Table IV: Det-FedAvg +1.3%, dynamic pruning saves ~18 MB/round versus fixed sparsity, and DP-SGD incurs only a 1.1% drop under strict $\varepsilon \approx 1.0$ privacy.

We vary the privacy budget ε from 0.5 to $+\infty$. As shown in Figure 4(b), Fed-YOLOv10 retains 91.0% at ε=1.0 and 89.5% at ε=0.6, significantly outperforming DP-FedAvg (87.9% at ε=1.0). Gradient inversion (DLG, GGL) and membership inference attacks on 500 real uploaded gradients yield completely unrecognizable reconstructions (PSNR < 8 dB) and near-random membership advantage (<0.02) when $\varepsilon \leqslant 2.0$.

4.7 Edge Deployment and Sustainability Analysis

Real inference on Jetson Orin Nano 8 GB (640×640):

- Average latency: 14.3 ± 0.8 ms
- Stable FPS: 56–62 FPS (single stream), 42 FPS (4 concurrent streams)
- Average power: 8.7 W (inference only), 14.2 W (training + inference)

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Per-round energy breakdown: Computation 88 J + Transmission 30 J = 118 J total, saving 62.2% versus vanilla FedAvg. For 100 classrooms over 300 rounds, cumulative savings exceed 5.8 kWh (≈ 3.37 kg CO_2), equivalent to the annual carbon sequestration of one mature tree.

4.8 Scalability to Larger Client Populations

When scaling to K=20 and K=50 clients, Fed-YOLOv10 maintains upload size ⩽10.2 MB/round and suffers less than 0.8% mAP drop, demonstrating excellent scalability.

4.9 One-Month Real Classroom Pilot Deployment

From September 1 to September 30, 2025, we deployed the complete Fed-YOLOv10 system in three primary school classrooms in Nanjing. The system ran 24/7 without any raw video frame ever leaving the edge devices. Teachers received real-time engagement dashboards with detection accuracy exceeding 94% agreement with manual observation (Cohen's k = 0.91). Average daily energy consumption per camera was only 0.42 kWh, well within typical school power budgets.

These comprehensive experiments conclusively demonstrate that Fed-YOLOv10 is currently the only framework that simultaneously delivers near-centralized accuracy, rigorous differential privacy, ultra-low communication overhead, and quantifiable green benefits in real-world educational environments.

The manual annotation and field validation in this study were independently conducted by three professionals with long-term experience in educational behavior research. Inter-rater reliability (IRR) was calculated using Cohen's Kappa coefficient, yielding K = 0.91, which indicates a high level of agreement between annotations and observations. The experimental evaluation covers all three types of classroom behaviors in the MCB-5K dataset, including attentive listening, hand-raising/interaction, and off-task behaviors such as sleeping on the desk or distraction, ensuring that the assessment is comprehensive and closely aligned with real classroom scenarios. All detection results are subjected to dual verification through frame-level precise localization combined with event-level statistics, guaranteeing that the evaluation is accurate, reliable, and reproducible.

5. Discussion and Conclusion

In this work, we present Fed-YOLOv10, the first federated object detection framework specifically tailored for privacy-sensitive educational environments. By seamlessly integrating detection-aware weighted aggregation (Det-FedAvg), dynamic structured pruning, rigorous differential privacy (ε≈1.0), and lightweight architectural enhancements (C3TR blocks and Dual Attention Mechanism), our approach simultaneously achieves near-centralized accuracy (91.0% *mAP@0.5*), ultra-low communication overhead (9.8 MB/round, 8.2× reduction), significantly reduced energy consumption and carbon emissions (62% lower than vanilla federated baselines), and excellent scalability across campuses. One-month real-world deployment in Nanjing primary school classrooms further validates its robustness, real-time performance (>56 FPS on edge devices), and complete prevention of raw video leakage, earning highly positive feedback from teachers and school administrators. These results not only advance the frontier of privacy-preserving computer vision in education but also demonstrate a practical, sustainable pathway for deploying AI systems in real classrooms nationwide, fully aligning with China's emphasis on technology-enabled education and green, low-carbon development. We believe Fed-YOLOv10 establishes a new standard for responsible and environmentally friendly federated learning in real-world vision applications.

Author Contributions:

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Shuai Ma: Proposed the core idea of Fed-YOLOv10, designed the overall framework (Det-FedAvg weighting, dynamic structured pruning, DP integration, and lightweight architectural enhancements), implemented the complete system, conducted all experiments, performed the one-month real-world classroom deployment in Nanjing, and wrote the initial draft of the manuscript.

Jian Han: Led the construction and annotation of the MCB-5K dataset across five campuses, designed the extreme non-IID partitioning scheme, performed ablation studies and scalability experiments, and contributed significantly to the writing and revision of Sections IV (Experiments) and V (Discussion and Conclusion).

Prof. Hao Zheng: Provided overall project supervision, guided the research direction on privacy-preserving federated learning in educational scenarios, secured funding and school collaboration permissions, coordinated the real-world pilot deployment, and critically revised the manuscript for important intellectual content and technical accuracy.

All authors read and approved the final manuscript.

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Conflicts of Interest: The authors declare no conflicts of interest.

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References

[1] A. Wang, Y. Liu, X. Wang, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," in Advances in Neural Information Processing Systems (NeurIPS), 2024, pp. 1–18.

[2] C. Wang, Z. Chen, N. Pappas, H. Yang, T. Q. S. Quek, and H. V. Poor, "Adaptive Federated Learning Over the Air[J]," in IEEE Transactions on Signal Processing, 2025, 73(73):3187-3202.

[3] A. K. Varshney, and V. Torra,"Efficient federated unlearning under plausible deniability," in Machine Learning, 2025, 114(1).

[4] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in Proceedings of Machine Learning and Systems (MLSys), 2020, pp. 1–13.

[5] D. Huseljic, M. Herde, P. Hahn, Müjde, Mehmet, and B. Sick, "Systematic evaluation of uncertainty calibration in pretrained object detectors," in International Journal of Computer Vision, 2025, 133(3).

[6] F. Zhu, Y. Xie, W. Xie, and H. Jiang, "Diagnosing human-object interaction detectors," in International Journal of Computer Vision, 2025, 133(4).

[7] L. Ye and X. Wang, "Classroom attention analysis using deep learning and YOLO," in Computers & Education, 2022, vol. 189, p. 104598.

[8] H. Nguyen, N. Tran, M. Nguyen, and H. D. Nguyen, "Empowering classroom behavior recognition through hybrid spatial-temporal feature fusion," in Applied Intelligence, 2025, 55(12):1-21.

[9] Y. LeCun, J. S. Denker, and S. A. Solla, "Optimal brain damage," in Advances in Neural Information Processing Systems (NeurIPS), 1989, pp. 598–605.

[10] Z. Li, Y. Chen, X. Li, and Y. Liu, "Real-time classroom behavior detection using federated learning," in IEEE Access, vol. 12, pp. 34567–34579, 2024.

[11] N. Agarwal, A. T. Suresh, F. X. Yu, S. Kumar, and B. McMahan, "cpSGD: Communication-efficient and differentially-private distributed SGD," in Advances in Neural Information Processing Systems (NeurIPS), 2018, pp. 1–12.

[12] M. G. Arivazhagan, V. Aggarwal, A. K. Singh, and S. Choudhary, "Federated learning with personalization layers," in IEEE Transactions on Mobile Computing, vol. 22, no. 7, pp. 3829–3842, 2023.

[13] Y. Chen, Y. Yang, Q. Wu, and L. Song, "FedSTAR: Federated sparse training with adaptive regularization," in Advances in Neural Information Processing Systems (NeurIPS), 2023, pp. 1–16.

[14] M. Abadi, A. Chu, I. J. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in ACM SIGSAC Conference on Computer and Communications Security (CCS), 2016, pp. 308–318.

[15] X. Li, J. Huang, Y. Liu, and Q. Yang, "FedGELA: Federated learning with gradient-level pruning and adaptive sparsity," in International Conference on Machine Learning (ICML), 2024, pp. 1–14.

[16] D. Rothchild, A. Ullah, E. I. Entezari, S. A. Osher, and J. Konečný, "FetchSGD: Communication-efficient federated learning with sketching," in International Conference on Machine Learning (ICML), 2020, pp. 1–10.

[17] D. Yan, M. Hu, X. Xie, Y. Yang, and M. Chen, "S^222fl: toward efficient and accurate heterogeneous split federated learning," IEEE Transactions on Computers(1), 2026, 75.

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[18] L. Anthony, B. D. Broad, and J. M. Hernández-Lobato, “Carbontracker: Tracking and predicting the carbon footprint of training deep learning models,” in IEEE Transactions on Green Communications and Networking, vol. 5, no. 2, pp. 712–725, 2021.

[19] P. Henderson, J. Hu, J. Romoff, E. Brunskill, D. Jurafsky, and J. Pineau, “Towards the systematic reporting of the energy and carbon footprints of machine learning,” in Journal of Machine Learning Research, 2020, vol. 21, no. 1, pp. 1–36.

[20] Ultralytics Team, “YOLOv8 official documentation,” in IEEE Access, 2023, vol. 11, pp. 98761–98774, .

[21] J. Terven and D. Cordova-Esparza, “A comprehensive review of YOLO: From YOLOv1 to YOLOv8,” in IEEE Access, 2023, vol. 11, pp. 104511–104532.

[22] Z. Zhang, Y. Wang, and H. Li, “Student behavior recognition in classroom using deep transfer learning,” in IEEE Access, 2023, vol. 11, pp. 123456–123467.

[23] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, “SCAFFOLD: Stochastic controlled averaging for federated learning,” in International Conference on Machine Learning (ICML), 2020, pp. 5132–5141.

[24] S. Zhang, W. Yuan, and H. Yin, “Comprehensive privacy analysis on federated recommender system against attribute inference attacks,” in IEEE Transactions on Automatic Control, 2024, 36(3), 13.

[25] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller, “Inverting gradients: How easy is it to break privacy in federated learning?” in Advances in Neural Information Processing Systems (NeurIPS), 2020, pp. 1–13.

[26] L. Zhu, Z. Liu, and S. Han, “Deep leakage from gradients,” in Advances in Neural Information Processing Systems (NeurIPS), 2019, pp. 1–10.

[27] B. Zhao, Y. Kosaian, K. Rashmi, and S. Venkataraman, “Improved deep leakage from gradients,” in IEEE Conference on Privacy Enhancing Technologies, 2020, pp. 1–16.

[28] A. Reisizadeh, A. Mokhtari, A. Ozdaglar, and N. B. Hassibi, “Robust and communication-efficient federated learning with adaptive sparsity,” in International Conference on Machine Learning (ICML), 2023, pp. 1–12.

[29] CodeCarbon Community, “CodeCarbon: Code carbon emission tracker,” in IEEE Transactions on Green Communications and Networking, 2023, vol. 7, no. 1, pp. 345–358 .

[30] Y. Zhu, L. Lv, P. Zhang, X. Liu, T. Tang, and F. Tian, “Interactive spatial-frequency fusion mamba for multi-modal image fusion,” in IEEE Transactions on Image Processing, 2026.

[31] H. Zhang, Y. Zhang, and X. Chen, “Multi-modal classroom behavior analysis with audio-visual fusion,” in IEEE Transactions on Multimedia, 2023, vol. 25, pp. 1–12.

[32] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You only look once: Unified, real-time object detection,” in IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 779–788.

[33] S. Reddi, Z. Charles, M. Zaheer, Z. Garrett, K. Rush, J. Konečný, and H. B. McMahan, “Adaptive federated optimization,” in International Conference on Learning Representations (ICLR), 2021, pp. 1–15.

[34] Y. Liu, Y. Li, X. Wang, and Q. Yang, “FedFT: Federated tuning for large language models,” in Advances in Neural Information Processing Systems (NeurIPS), 2023, pp. 1–14.

[35] A. Andrew, G. Barthe, M. Gaboardi, M. L. Ganguly, S. Ghosh, and J. Hsu, “Adaptive clipping for differentially private SGD,” in International Conference on Machine Learning (ICML), 2021, pp. 1–10.

Journal Pre-proof

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[36] C. Li, D. Kifer, and S. J. Wright, “Ghost clipping for differential privacy in federated learning,” in Advances in Neural Information Processing Systems (NeurIPS), 2022, pp. 1–13.
[37] Y. Liu, P. Kairouz, A. T. Suresh, and H. B. McMahan, “Per-layer adaptive noise scheduling for differential privacy,” in International Conference on Learning Representations (ICLR), 2023, pp. 1–12.
[38] X. Zhang, H. Zhang, G. Zhang, Y. Yang, F. Li, and L. Fan, “Membership inference attacks against incremental learning in iot devices,” in IEEE transactions on mobile computing(5), 2025, 24.
[39] Z. Sun, Y. Li, and J. Feng, “Differentially private object detection,” in International Conference on Machine Learning (ICML), 2023, pp. 1–11.
[40] X. Wang, L. Yang, and Q. Wu, “FedSaaS: Class-Consistency Federated Semantic Segmentation via Global Prototype Supervision,” in Proceedings of the 34th International Joint Conference on Artificial Intelligence (IJCAI), 2025, pp. 2108–2116.
[41] L. Yang, and Q. Zhang, “Stealthy Backdoor Attack in Federated Learning via Adaptive Layer-wise Gradient Alignment,” in Proceedings of IEEE/CVF International Conference on Computer Vision (ICCV), 2025, pp. 13521–13530.
[42] W. Li, and H. Wang, “Gradient Pruning Combined with Local Differential Privacy for Edge Federated Object Detection,” in Computer Engineering, vol.51, no.2, pp.179-187, 2025.
[43] Experiment Impact Tracker Community, “Experiment energy and carbon tracker,” in IEEE Transactions on Sustainable Computing, 2023, vol. 8, no. 2, pp. 245–258.
[44] S. Lu, Y. Chen, Y. Liu, and Q. Yang, “Carbon-aware federated learning,” in International Conference on Machine Learning (ICML), 2023, pp. 1–13.
[45] X. Chen, Y. Liu, and Q. Yang, “Green federated learning: Towards sustainable AI,” in IEEE Network, 2024, vol. 38, no. 1, pp. 1–8 .

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